

VIJAYAKUMAR K

Full-Stack & Application Developer | Freelance Software Developer

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[GitHub](#) | [LinkedIn](#)

"I don't just build websites. I build complete software products."

Full-Stack & Application Developer building production-focused web platforms, Android applications, desktop software, business systems, and AI-powered tools. 2+ years of hands-on project experience across the MERN stack, Kotlin/Jetpack Compose, Electron, and Python — taking products from idea, through interface design, to real production deployment. Comfortable owning a project end-to-end: requirements, architecture, database design, RESTful APIs, authentication/RBAC, UI build, deployment, and client handover.

CORE CAPABILITIES

Web Applications

React, Vite, MUI, Tailwind, Node.js, Express, MongoDB

Production-ready, multi-role web portals, ERPs, and complex administrative SaaS platforms built with modular frontend architecture and robust REST APIs.

Android Development

Kotlin, Jetpack Compose, Firebase, Room, Mapbox

Modern native Android applications featuring real-time offline sync, spatial Mapbox navigation, hardware integrations, and Tamil text-to-speech.

Desktop Applications

Electron, React, SQLite, native packaging and local workflows

Cross-platform Windows desktop applications with local-first databases, dot-matrix hardware printing, native installer builds (NSIS), and background workers.

AI & Automation

Gemini AI, OCR, document processing, translation and automation

Intelligent workflow enhancements combining LLMs, computer vision, local OCR engines, and automated document conversion to replace repetitive tasks.

SERVICES OFFERED

Custom Web Applications — Admin dashboards, business portals, multi-role workflow platforms, and complete management systems engineered with React, Node.js, and MongoDB.

Android Applications — Modern native Android applications built using Kotlin and Jetpack Compose, featuring Mapbox spatial navigation, Room offline sync, and Firebase.

Desktop Software — Cross-platform Windows desktop software using Electron and React, designed for offline hardware printing, local SQLite databases, and business billing.

Business Automation & AI — Replace repetitive manual operations with automated document parsing, OCR pipelines, Gemini AI integration, font encoding converters, and translation engines.

Existing System Improvements — Modernize outdated user interfaces, add new operational features, fix performance bottlenecks, restructure databases, and upgrade application workflows.

TRACK RECORD

40+	13	40+	3
Screens in a single platform (EasyAudit)	User roles handled (MGBuilders ERP)	Backend route controllers (flagship builds)	Platforms: Web, Android, Desktop

FEATURED PROJECTS

MGBuilders ERP — Construction Material ERP System - [Live Demo](#)

Web Platform | React, Vite, Node.js, Express, MongoDB, Mongoose, Tailwind CSS

A comprehensive role-driven ERP platform managing the complete construction-material lifecycle from request and procurement through stock movement and reporting.

- 25+ application pages with complex relational views
- 13 granular user roles with dynamic route permissions
- End-to-end workflow: Request → Purchase Order → Receipt → Stock → Issue → Return → Transfer
- Daily stock snapshot architecture for instant ledger auditing
- Fiscal-year serial numbering & automated document generation
- Export to styled PDF and multi-sheet Excel reports

EasyAudit — Audit Workflow & Document Tracking Platform - [Live Demo](#)

Web Platform | React, Vite, Material UI, Node.js, Express, MongoDB

A multi-role audit management platform organizing client registration, tax assessments, notices, communication, documents, and complete activity history.

- 40+ responsive screens for accounting & auditing professionals
- Multi-role access hierarchy: Admin / SuperUser / AuditUser
- Tax assessment tracking and IT notice filing management
- Integrated in-app messaging and secure client file uploads
- 42 RESTful API endpoints powering client data workflows

Employee Management System — Role-Based Task, Leave & Communication Platform - [Live Demo](#)

Web Platform | React 18, Vite 6, MUI 6, Redux Toolkit, Socket.io, Node.js, Express, MongoDB, JWT

A full-stack enterprise application with role-based interfaces for Administrators and Employees to manage records, delegate tasks, process leave, and communicate in real time.

- Dual role-based interfaces (Administrator vs. Employee dashboard)
- Task delegation & progress tracking with automated notifications
- Real-time notifications powered by Socket.io
- Interactive analytics with Chart.js and Recharts
- JWT authentication & bcryptjs password hashing

FuelStationBillingApp — Fuel Station Billing & Management System

Windows Desktop | Electron, React, Node.js, Express, MongoDB, NSIS

Desktop billing and operations software designed around real fuel-station workflows.

- GST invoice generation with dynamic tax split (CGST / SGST / IGST)
- Pump nozzle opening & closing meter reading tracking
- Shift 1 / Shift 2 workflow and teller reconciliation
- Hardware integration for dot-matrix impact slip printers
- Automated Google Drive cloud database backup

FaceHelp / CCTVHelper — Offline CCTV Face Recognition System

Windows Desktop | Electron, React, face-api.js, HLS Video, HTML5 Canvas

Privacy-oriented desktop software for searching and monitoring CCTV footage using local face-recognition processing.

- 100% local/offline video stream scanning with privacy enforcement
- Real-time facial detection powered by face-api.js
- Target Search mode and Free Roam face clustering
- Configurable AI match confidence thresholds

GK Studio / BigWin — Tamil Document Processing & AI Studio
Windows Desktop | Electron, React, Gemini AI, Tesseract.js, PaddleOCR, Python

Desktop productivity software combining Tamil encoding conversion, translation, OCR, and generative AI document workflows.

- Legacy Bamini ↔ Unicode text encoding conversion engine
- High-accuracy English ↔ Tamil machine translation
- Multi-engine OCR pipeline (Tesseract.js + Python PaddleOCR)
- Generative AI document enhancement via Gemini API

KLS System — Analytics & AI Data Platform
Windows Desktop | Electron, React, SQLite, better-sqlite3, Gemini AI

Data analytics & software engineering platform featuring historical trend frequency engines, OCR scanning, and a database-aware AI assistant.

- Embedded SQLite database with native better-sqlite3 bindings
- Frequency distribution engines and historical pattern metrics
- OCR scanner to extract data from uploaded images
- Database-aware Gemini AI assistant with direct SQL query context

ANDROID PROJECTS

Business Map — Parking / Business Discovery & Navigation
Kotlin, Jetpack Compose, MVVM, Mapbox Maps, Firebase Firestore, Room

- MVVM clean architecture with Jetpack Compose declarative UI
- Mapbox SDK integration for custom vector maps & turn-by-turn navigation
- Firestore ↔ Room offline-first database synchronization

FormMap — Multi-Step Business Registration
Kotlin, Jetpack Compose, Firebase Firestore, GPS Location

- Multi-step registration workflow with GPS coordinate autofill
- Dynamic custom form fields with real-time validation
- Searchable business directory with instant filtering

Receptionist — Hospital Token / Queue Management
Kotlin, Jetpack Compose, Firebase Realtime DB, Tamil TTS

- Real-time daily token generation and FIFO queue orchestration
- Tamil Text-to-Speech integration for patient announcements
- Returning-patient lookup and missed-call queue reordering

StoreKeeper — Supermarket Inventory & QR System
Kotlin, Jetpack Compose, Firebase, CameraX / QR

- High-speed QR code generation & scanning using CameraX API
- Hierarchical storage model: Shelf → Rack → Item
- On-device image compression prior to cloud storage

TECHNICAL EXPERTISE

Frontend	React, Vite, JavaScript, Material UI, Tailwind CSS, Framer Motion, Redux Toolkit, Recharts
Backend	Node.js, Express, REST APIs, MongoDB, Mongoose, JWT Auth, Session Auth, Role Permissions
Android	Kotlin, Jetpack Compose, Material 3, Room Database, Firebase, Mapbox Maps, CameraX, Tamil TTS
Desktop	Electron, SQLite, IPC Communications, NSIS Packaging, Native Modules, Dot-Matrix Printing
AI & Automation	Gemini AI, OCR, Tesseract.js, PaddleOCR, Python Subprocess, Document Automation, Translation
Additional Engineering	RBAC Access Control, Dynamic Permissions, PDF/Excel Generation, Google Drive API, Realtime DB

HOW I WORK

- 01 Discover** — Deep dive into business requirements, user roles, operational bottlenecks, and technical constraints.
- 02 Architect** — Design clean data schemas, application routing permissions, state management, and UI component hierarchy.
- 03 Build** — Develop robust frontend interfaces, scalable backend APIs, database models, and platform-native logic.
- 04 Refine** — Rigorous testing across roles, mobile responsiveness polish, keyboard navigation, and performance optimization.
- 05 Deliver** — Deploy web platforms to cloud servers or package desktop NSIS installers and Android APK builds.

PROFESSIONAL BACKGROUND

Independent Software Developer — Hands-on Project Experience

Feb 2024 – Present

- Engineered production-ready full-stack applications from concept to deployment using reusable, modular UI components.
- Designed and integrated secure RESTful APIs for cross-platform data synchronization and reliable data consistency.
- Implemented JWT/OAuth-based authentication and Role-Based Access Control (RBAC) for secure application access.

Office Staff & Operations Support — Safexpress Pvt. Ltd.

2018 – 2023

- Managed office administration, daily billing, invoice processing, and logistics/transport documentation.
- Maintained accurate data entry and tracking systems, improving record-keeping reliability.

Document Writer Assistant — TNREGINET & Registration Services

2023 – Present

- Assist with document preparation and TNREGINET online services, ensuring accurate data entry and documentation.

EDUCATION & CERTIFICATIONS

- Bachelor of Commerce (B.Com), Cardamom Planters Association College, Tamil Nadu — 2014 – 2017
- Java Programming Certification (6-Month Course), CAD CAE Computers — 2023